ASUS UGEN AI LEAGUE INNOVATION HACKATHON 2026 · STAGE I PROPOSAL

Spec to Inspector

The customer's own defect specification, enforced and cited on every reject.

REJECT

Clause 5.5 Damaged component or board page 5 of the specification

Region 1: clause 5.5 rejects 'damage' in any zone, whatever its size.

A reject from the laptop prototype: the clause, its page in the specification, and the rule's own sentence. Code decides; the language model only names the defect and the clause.

Theme and track	Workplace AI · Battlefield Thunderstorm (ASUS UGen Intel Arc Pro B70 32GB) · Global Online
Team	Brian Isaac (solo)
Code	https://github.com/BrianIsaac/spec-to-inspector
Demo video	added before submission

Screen: docs/screens/page-01-reject-clause-line.png, the laptop prototype's console on VisA PCB4 test board 018, 11 October 2026. The specification it cites is a demonstration document written for this entry (appendix A).

AT A GLANCE

One specification in, one inspector out

Every decision is traceable to a clause of the customer's own document. Built from parts Intel's own stack lists for the Arc Pro B70, and measured end to end on a laptop for Stage I.

- **The problem.** A supplier's quality engineer turns each customer's defect specification into inspection rules by hand, and a reject that cannot name its clause is argued from memory.
- **What it does.** Parses the specification into typed clauses; an anomaly model finds the defect; code measures it; a vision-language model names the defect and the clause; a rule applies the clause's limit; the engineer's correction is kept against the clause.
- **Why one B70.** Its 32 GB holds an 8B vision-language model at FP16 or FP8, its context and the anomaly model together, on the supplier's own premises, where a customer's confidential specification stays.

188 of 200
defective boards rejected (174) or referred to an engineer (14)

38.5%
right clause cited: 52 of 135 single-defect boards (95% interval 30.7% to 46.9%)

16 of 201
good boards wrongly rejected (8.0%)

12.3 min
from a new specification to a working inspector, one part

Development figures: the thresholds were chosen on the same test split. The clause figure is low and is printed as measured; page 12 has the table and page 17 what Stage II must change.

JUDGING CRITERION (WEIGHT)	PAGES
Innovation and Creativity (30%)	4, 5, 15
Commercial Potential and Feasibility (35%)	12, 13, 14, 16, 17, 18, 19
Practical Value (25%)	3, 6, 11, 18
Storytelling and Presentation (10%)	the whole deck; the video; page 6

WHAT THE CONTEST ASKS THE DECK TO HOLD	PAGES
Scenario and problem	3, 4
Tools and methods	5 to 11
Expected outcomes	12, 17
Hardware and software architecture	13, 14, 16
References and the GitHub link	20; the link also on page 1

Code and its documentation: https://github.com/BrianIsaac/spec-to-inspector

Figures: results/stage1/bench.json, written by `uv run spec-to-inspector bench` on 11 October 2026 (laptop: RTX 5070 Ti, 12 GB; VLM at 4-bit); method in appendix B. Intel's list for the B70: llm-scaler's model table, reference 11. Criteria and weights: the contest's page, reference 1, read 11 October 2026.

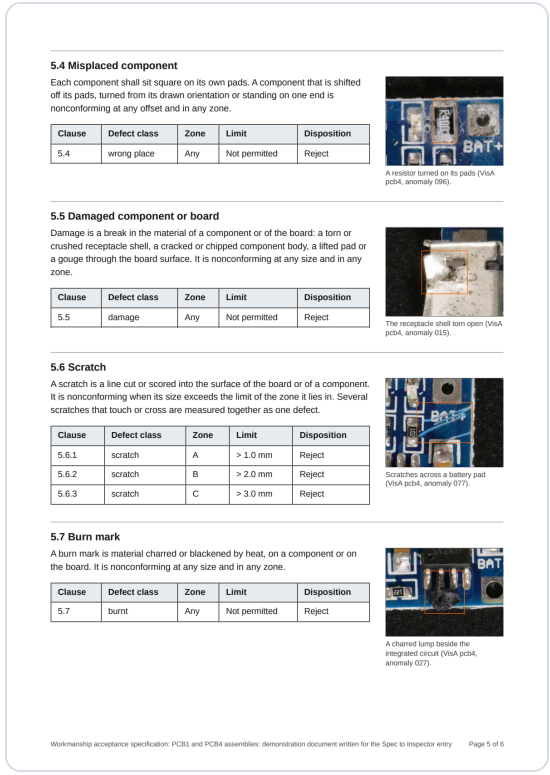
The engineer's problem: which clause did this board fail?

A supplier's quality engineer turns a customer's defect specification into inspection rules by hand, and loses every argument about a reject that cannot name its clause.

> "The customer sends a 30-page defect specification with photos. It takes us weeks to turn it into inspection rules. When we reject a part, nobody can say which clause it failed, and when they reject ours, we argue from memory."
>
> A quality engineer at a component supplier: a persona written for this proposal, not a quotation from a person.

What it costs, in words:

- **Delay.** A new part cannot be inspected until someone has read the document and written its rules.
- **Drift.** The rules live in an inspection recipe, the limits in a document; after the next revision they no longer say the same thing.
- **Disputes.** A reject that names no clause is settled by whoever remembers the specification better.



5.4 Misplaced component

Each component shall sit square on its own pads. A component that is shifted off its pads, turned from its drawn orientation or standing on one end is nonconforming at any offset and in any zone.

Clause	Defect class	Zone	Limit	Disposition
5.4	wrong place	Any	Not permitted	Reject

A resistor turned on its pads (VisA pcb4, anomaly 096).

5.5 Damaged component or board

Damage is a break in the material of a component or of the board: a torn or crushed receptacle shell, a cracked or chipped component body, a lifted pad or a gouge through the board surface. It is nonconforming at any size and in any zone.

Clause	Defect class	Zone	Limit	Disposition
5.5	damage	Any	Not permitted	Reject

The receptacle shell torn open (VisA pcb4, anomaly 015).

5.6 Scratch

A scratch is a line cut or scored into the surface of the board or of a component. It is nonconforming when its size exceeds the limit of the zone it lies in. Several scratches that touch or cross are measured together as one defect.

Clause	Defect class	Zone	Limit	Disposition
5.6.1	scratch	A	> 1.0 mm	Reject
5.6.2	scratch	B	> 2.0 mm	Reject
5.6.3	scratch	C	> 3.0 mm	Reject

Scratches across a battery pad (VisA pcb4, anomaly 077).

5.7 Burn mark

A burn mark is material charred or blackened by heat, on a component or on the board. It is nonconforming at any size and in any zone.

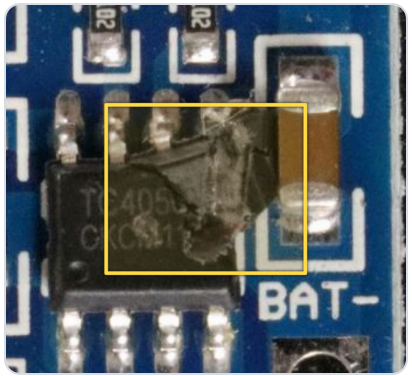
Clause	Defect class	Zone	Limit	Disposition
5.7	burnt	Any	Not permitted	Reject

A charred lump beside the integrated circuit (VisA pcb4, anomaly 027).

Workmanship acceptance specification: PCB1 and PCB4 assemblies: demonstration document written for the Spec to Inspector entry Page 5 of 6

?

which clause



Left: page 5 of the demonstration specification, specs/pcb-spec.pdf, written for this entry in the style of a customer's document. Right: VisA PCB4 test board 018 as the console shows it (docs/screens/page-05-2-finds.png); VisA is Amazon's dataset, CC BY 4.0, reference 6. No market-size figure is printed: none was found at a source.

What exists, and the step nobody ships

Detection, explanation and criteria typed into a prompt all exist. In the sources we read to 11 October 2026 we found nothing that takes in the customer's own specification and quotes its clause and limit on every reject.

NEAREST NEIGHBOUR	WHAT IT DOES, IN ITS OWN WORDS	WHERE IT STOPS
Intel Open Edge Platform 2026.2, 10 September 2026	"defect detection into actionable quality insights through agentic workflows, VLM-based explainability"	Explains a defect; no customer document is read and no clause is cited.
Intel with Advantech, Pro Day fact sheet, March 2026	On Arc Pro B-Series: a VLM with retrieval "to help engineers analyze defects, reference documentation, and identify likely root causes"	Retrieves documentation for an engineer to read; binds no accept or reject to a clause and its limit.
PCBA Standard-to-Real Challenge (ASUS and NTU), ACM Multimedia 2026	"cross-domain PCBA inspection VQA … defect identification, quantitative reasoning, and defect cause/treatment"	Answers questions, scored as a benchmark. Ours is the step after: a customer's own contract specification, cited on each decision.
GLLS, arXiv 2607.03817, 4 July 2026	"structured specifications in the inference context"; decisions "traceable to explicit visual evidence"	Traceable to visual evidence, not to a clause of a customer's document.
TunnelMIND, arXiv 2604.27928, 30 April 2026	Measures a defect, grades severity "under expert knowledge constraints", writes a "retrieval-grounded explanation"	Tunnels; thresholds set by hand, not parsed from a customer's document; no memory of overrides.
Aqrose AIDI 3.4, 17 July 2026; VISION 2026 notice, 3 September 2026	"Just describe your needs in natural language"; an AI Review for "consistency, traceability and decision efficiency"	Needs are typed by a person; no document is read and no clause cited.
Overview AI OV Spark, 6 October 2026	"Describe your part and what should pass or fail."	A setup assistant on the camera; its record is a capture library and an audit log, not a clause.
ZEDEDA, 19 March 2026 (older: context)	A fixed 8B VLM on Jetson Thor; the inspection policy is "a natural language prompt"	The rules are a prompt an engineer writes.

Not claimed as new, because each exists: anomaly detection from good images (Anomalib); a VLM explaining a defect (Intel); corrections fed back; answers that cite a source document; and "measure, a rule decides, a model explains" (TunnelMIND does it for tunnels). The incumbent vision suites (MVTec, Keyence, Cognex, Landing AI): nothing found that reads a specification or cites a clause.

Claimed: the narrow chain. The customer's PDF becomes typed clauses; code measures; the model names the clause; code applies its limit; the reject quotes clause and limit; the override is kept per clause.



Each row's source, with its URL, is on page 20: Intel references 14 and 15; the challenge 16; GLLS 17; TunnelMIND 19; Aqrose 20; Overview 21; ZEDEDA 22. Quotations were read at those pages on 11 October 2026; the incumbents' 2026 releases were searched on 9 October 2026 and not every page could be opened. Simplified-Chinese platform vendors were not searched.

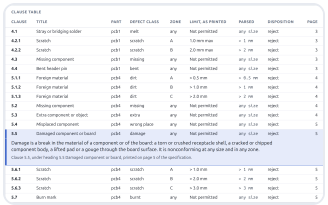
The specification is the source of truth

The anomaly model finds, code measures, the vision-language model names the defect and the clause, a rule decides, and the engineer corrects. No number comes from the language model.

1 · PARSE

The specification becomes clauses

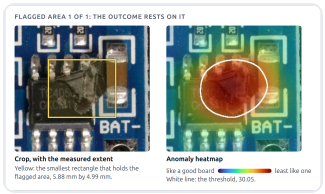
The customer's PDF is parsed into a typed table: clause, defect class, zone, limit, page.



2 · FIND

The anomaly model finds

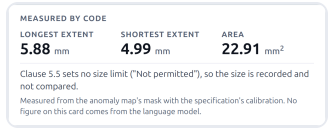
Trained on good boards only, it flags what differs from good and where.



3 · MEASURE

Code measures

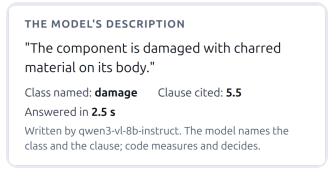
The flagged area's extent in millimetres and its zone, from the specification's own calibration.



4 · NAME

The model names

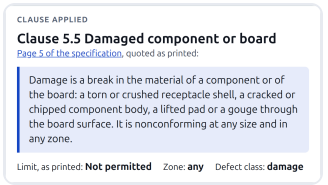
A vision-language model reads the crop and the clause table and names a defect class and a clause.



5 · DECIDE

A rule decides and cites

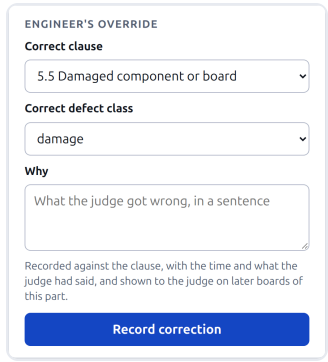
Code applies the clause's limit to the measurement and quotes the clause on the reject.



6 · CORRECT

The engineer corrects

An override is stored against the clause with its reason, and shown to the judge on later boards.

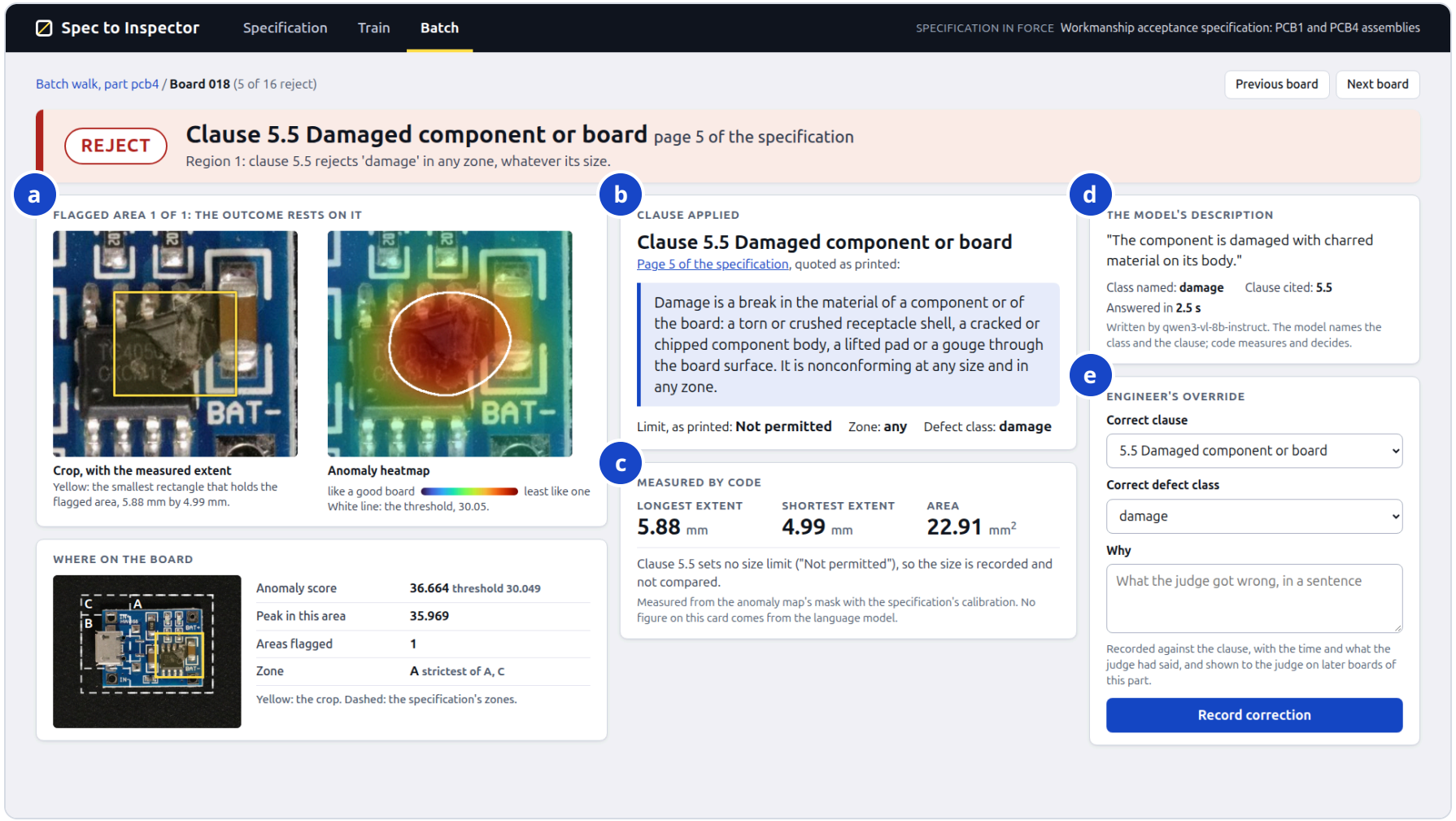


The customer's own specification, cited on every reject.

Screens: docs/screens/page-05-1-specification.png to page-05-6-corrects.png, the console's own panel for each step, on VisA PCB4 test board 018 (11 October 2026). The specification is the demonstration document of appendix A.

One reject, end to end

A reject that shows its grounds: the crop, the clause as the customer printed it, the size measured by code, and a way to correct it.



- **a** **The crop and the heatmap.** Where the anomaly model flagged, with the measured rectangle drawn on the photograph.
- **b** **The clause, quoted as printed**, with its page in the specification, its limit, zone and defect class.
- **c** **Measured by code** from the anomaly map's mask. This clause sets no size limit, so the size is recorded and not compared.
- **d** **The model's one line.** It names the class and the clause and writes no figure.
- **e** **The override.** The engineer picks the right clause and says why.

Screen: docs/screens/page-06-reject-card.png, the console on VisA PCB4 test board 018, 11 October 2026. The board was chosen by rule, not by eye: the first reject of a 24-board batch on which the judge named VisA's own defect type. How often it does is on page 12. The answer time on the card is that run's, not the bench's.

Method 1: the specification becomes clauses

A born-digital specification parses to a typed clause table without anyone retyping it: 6 pages, 16 clauses under 11 numbered headings and 5 zones, in 7.9 s on the CPU.

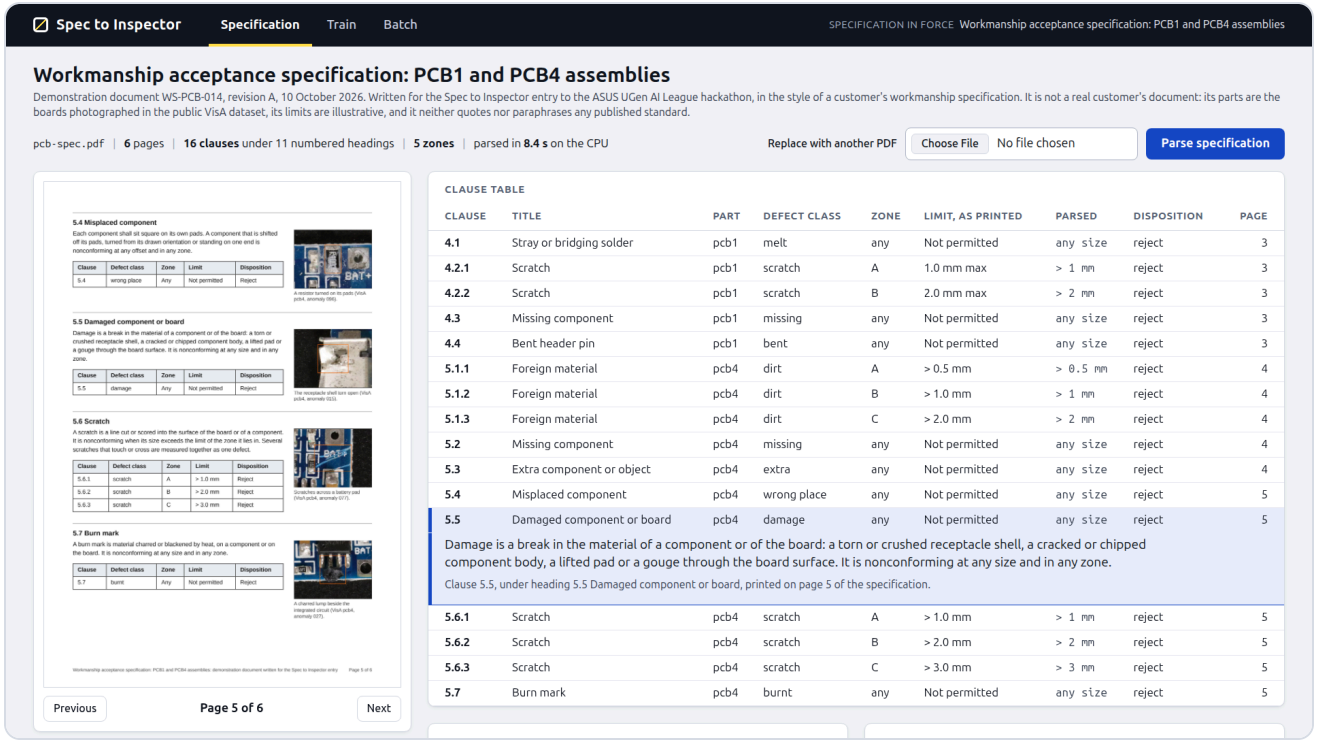
The clause, as a record: id and heading; title; part; defect class; zone; the limit as printed and as a number with its unit and comparison; disposition; page; the source text.

The parser: docling, with heading levels kept so that 5.1.2 stays under 5.1. Tables of limits are read as numbers, not prose: "1.0 mm max" and "> 1.0 mm" both become `> 1 mm`.

What is checked: every clause recovered and every limit parsed with its unit. The bench refuses to run unless today's parse equals the tracked clause table.

The calibration and the zones are declared in the specification too, so a size in millimetres and a zone are the customer's, not ours.

Scanned specifications go through an OCR vision-language model on the card: STAGE II PLAN



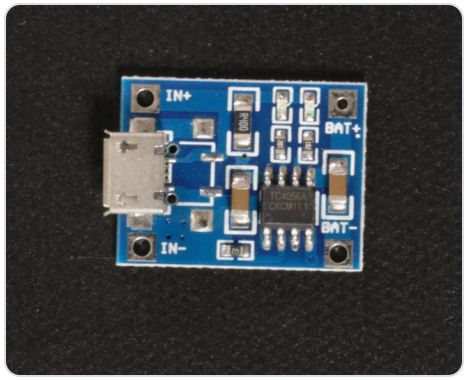
The PDF's page on the left, the parsed table on the right; clause 5.5 is open, quoted with its page.

Screen: docs/screens/page-07-specification-clause.png. Counts: results/stage1/spec.json (10 October 2026). Parse time: results/stage1/bench.json (11 October 2026); the run on screen took 8.4 s. docling 2.135.0, MIT: reference 9. The scanned-page path is planned for Stage II and is not in the Stage I prototype.

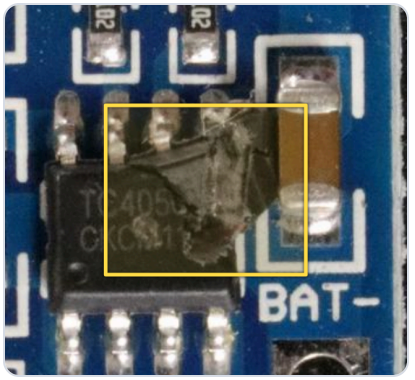
Method 2: the anomaly model learns "good" from good parts

The inspector needs only good images to start; a defect is what differs from good.

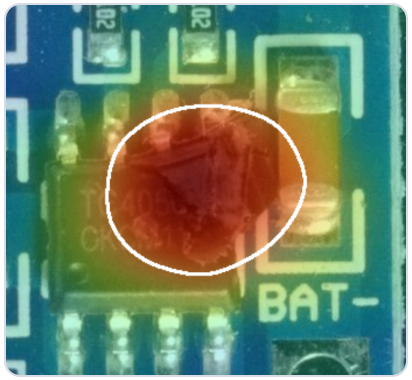
- **Anomalib 2.7.0** (Intel, Apache-2.0), PatchCore with its default `wide_resnet50_2` backbone, trained on the 904 good images of a VisA PCB part at 256 px. One fit takes 12.2 minutes on the laptop.
- **Why the anomaly model finds and the language model does not.** The VAND 4.0 challenge report (3 October 2026): "the best off-the-shelf VLM approach trails specialized models by ~28 AP, confirming that currently VLMs cannot replace fine-tuned detectors".
- **What it gives the next step:** a score for the board, a map of where it differs, and the flagged area's outline.
- **A new part with only 10 to 50 good images** is the few-shot case: STAGE II PLAN to measure, not measured in Stage I.



A good board: all the model is trained on



A test board: the flagged area



Its anomaly map; white: the threshold

IMAGE AUROC, ONE RUN PER PART	PCB1	PCB2	PCB3	PCB4
This prototype: PatchCore, `wide_resnet50_2`, 256 px	93.0	92.3	92.1	98.7
The Anomalib team's published ResNet18 figure, 256 px	94.9	92.4	99.0	99.0
Difference, in points	-1.9	-0.1	-6.9	-0.3

Not like for like: the published figure is the mean of three runs with another backbone. PCB3 is the cell we did not come near: every configuration we tried stayed more than five points under it. The judge and the clause figures use PCB1 and PCB4.

Images: VisA PCB4 (Amazon, CC BY 4.0) as the console shows them: good board 0107 from docs/screens/extra-accepted-card.png, test board 018 from page-05-2-finds.png. Training figures and AUROC: results/stage1/anomaly.json (10 October 2026); the published figure is Rolih et al., reference 5, its column 'Single model 256'. The VAND 4.0 report: reference 18, read 11 October 2026.

Method 3: code measures; the model does not guess numbers

Every size on a reject is measured from the anomaly map's mask in calibrated millimetres, and accept or reject is the clause's limit against that measurement, decided by code.

- **Calibration** is declared per part in the specification: PCB4's image covers 46.7 mm across. The scale is the customer's statement, printed on the card.
- **Size** is the long side of the smallest rectangle, at any angle, that holds the flagged area: the specification's own definition.
- **Zone** comes from the specification's zone map; where zones overlap, the most critical applies.
- **The limit check is a rule.** A size over the clause's limit rejects; "not permitted" rejects at any size. An area outside every zone is referred to an engineer, and so is an area with no outline when its clause sets a size limit: neither is guessed.
- **What it can resolve.** The anomaly map is 256 px across, 0.18 mm a pixel on PCB4, and it lights the neighbourhood of a defect, so a measured extent is often larger than the defect a person sees. A size near a limit is a case for the engineer; Stage II measures at a higher map resolution.

MEASURED BY CODE

LONGEST EXTENT	SHORTEST EXTENT	AREA
5.88 mm	**4.99** mm	**22.91** mm²

Clause 5.5 sets no size limit ("Not permitted"), so the size is recorded and not compared.

Measured from the anomaly map's mask with the specification's calibration. No figure on this card comes from the language model.

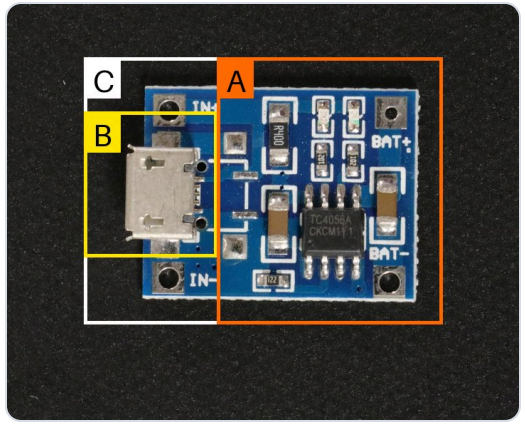
CLAUSE APPLIED

Clause 5.5 Damaged component or board

Page 5 of the specification, quoted as printed:

> Damage is a break in the material of a component or of the board: a torn or crushed receptacle shell, a cracked or chipped component body, a lifted pad or a gouge through the board surface. It is nonconforming at any size and in any zone.

Limit, as printed: **Not permitted** Zone: **any** Defect class: **damage**



The specification's zones for PCB4: A, components; B, the receptacle; C, pads and margin

Screens: docs/screens/page-05-3-measures.png and page-05-5-cites.png (board 018). Zones: specs/figures/pcb4-zones.jpg, from the demonstration specification. Calibration: specs/pcb-spec.clauses.json; map size: results/stage1/anomaly.json. The map's pixel is their quotient, not a measured resolution.

Method 4: the judge names the defect and the clause

A vision-language model reads the crop with the whole clause table in context and returns a structured answer; a rule then decides from the clause's limit and the measured size.

What the model is shown, in this order

1. The instructions and the whole clause table, each clause with its source text: 2,260 tokens, the same for every board, so the server caches it.
2. The part and its numbered headings.
3. Up to three engineer's corrections recalled for this part.
4. The measured facts, as JSON: the code's numbers, not the model's.
5. The crop, then the same crop under the anomaly heatmap.

What keeps it honest

- The answer is constrained to a JSON schema; its one line may hold no digit.
- A clause that is not in the table, or a class that is not that clause's, makes the board "needs review".
- The model may say that no clause applies; that too is "needs review". It can never accept a board.

Its answer for board 018

```
{
  "reason": "The component is damaged with charred material on its body.",
  "defect_class": "damage",
  "clause_id": "5.5"
}
```

The rule's sentence, written by code

```
clause 5.5 rejects 'damage' in any zone, whatever its size
```

Measured on the laptop: a prompt of 4,683 tokens at the median, an answer of 39 tokens in 3.33 s asked cold (2.55 s with the clause table cached). The count of bench citations whose class was not the cited clause's own: 0. As a contract it holds; as a classifier at 4-bit it is weak, and page 12 says how weak.

Prompt order: src/spec_to_inspector/judge/prompt.py. The answer shown is board 018's (docs/screens/page-05-4-names.png). Token counts and times: results/stage1/judge.json (11 October 2026) and bench.json (11 October 2026); laptop, llama.cpp, Qwen3VL-8B-Instruct-Q4_K_M.gguf. Qwen3-VL-8B-Instruct, Apache-2.0: reference 7.

Method 5: the engineer's correction is kept

The engineer's correction is stored, shown to the judge, and a similar defect on another board then cites the corrected clause. Every override is logged against its clause: the audit trail a customer asks for.

- **Stored:** the board and area, what the judge had said, the corrected class and clause, the engineer's note, the time. The judge's own decision is kept beside it, not overwritten.
- **Recalled:** the part's most recent corrections, three at most, in the judge's prompt.
- **Measured:** in 7 of 7 scripted pairs the second board, judged wrong before, cites the corrected clause after one override.
- **And its limit, measured too:** the same single correction also moved 14 of 19 control boards that truly are the class corrected away from; all 10 of 10 controls of the class corrected to held. With this model a correction acts as a rule about the class, not as an example.
- **So we do not claim** that corrections raise accuracy or teach the judge to tell two defects apart. No retraining happens in Stage I.

THE MODEL'S DESCRIPTION

"The USB receptacle has foreign material on its surface."

Class named: **dirt** Clause cited: **5.1**

Answered in **2.6 s**

Written by qwen3-vl-8b-instruct. The model names the class and the clause; code measures and decides.

Board 012 before the correction of board 005

THE MODEL'S DESCRIPTION

"The receptacle component is damaged with visible chipping and deformation."

Class named: **damage** Clause cited: **5.5**

Answered in **2.6 s**

Written by qwen3-vl-8b-instruct, with the engineers' corrections in view. The model names the class and the clause; code measures and decides.

Board 012 judged again afterwards

CORRECTIONS THE JUDGE WAS SHOWN

1. On a pcb4 board the judge called the flagged area 'dirt' (clause 5.1); the engineer corrected it to 'damage' (clause 5.5). The engineer's note: VisA labels this defect 'damage'; the judge named 'dirt'.

Recalled from the corrections store and put in the prompt with this crop.

The correction, as the judge was shown it and as the card prints it

Screens: docs/screens/page-11-3-second-board-before-model.png, page-11-4-second-board-after-model.png and page-11-4-second-board-after-shown.png: boards 005 and 012. The note on screen was written by the walk's script from VisA's label, not by an engineer. Counts: results/stage1/corrections.json (11 October 2026), each pair asked cold.

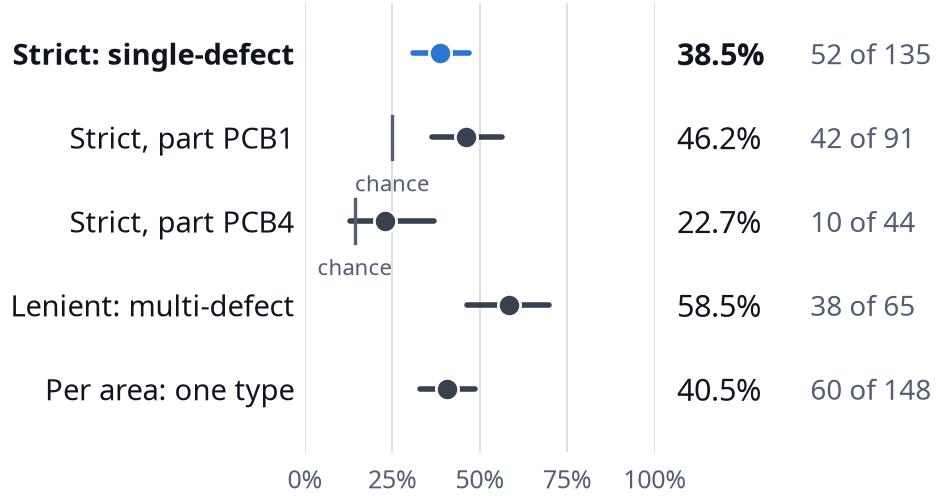
Stage I: what the laptop prototype measured

The pipeline runs end to end today on a 12 GB laptop GPU. These are its figures, the low one included.

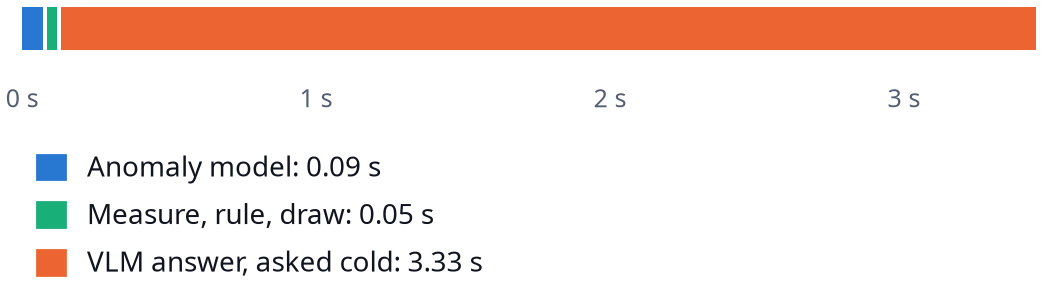
FIGURE	MEASURED ON 200 DEFECTIVE AND 201 GOOD VISA TEST BOARDS, PARTS PCB1 AND PCB4
Right clause cited, strict	**52 of 135 single-defect boards, 38.5%** (95% interval 30.7% to 46.9%). PCB1 42 of 91; PCB4 10 of 44
Defective boards stopped	188 of 200: 174 rejected, 14 referred to an engineer. The 12 that passed were not flagged by the anomaly model
Good boards wrongly rejected	16 of 201, 8.0% (5.0% to 12.5%); 9 more referred
A flagged board, end to end	3.49 s median, 4.03 s p95 (216 boards). A board that is not flagged: 86 ms
The VLM's answer, asked cold	3.33 s median, 3.84 s p95, at a prompt of 4,683 tokens (218 answers)
Anomaly model alone	20.3 to 20.6 images a second, from file to score
Peak GPU memory	judge 8,780 MiB; anomaly model while scoring 2,028 MiB; training 8,003 MiB
Specification to working inspector	12.3 min for one part: 7.9 s to parse, the rest training
Run twice	the same answer on 401 of 401 boards

How to read it. The clause is scored on the most anomalous area, with an empty corrections store; a judge citing at random would be right on 25.0% of PCB1's boards and 14.3% of PCB4's. The anomaly model is not what holds the figure down: on 109 of the 135 boards its area lies on the labelled defect, and the judge cites the right clause on 48 of those (44.0%).

Development figures. The thresholds were chosen on the same test split and the prompt was written with sight of some of its images. Labels are VisA's own defect annotation mapped to clauses by a written table, 20 spot-checked by hand (20 of 20 agree). The laptop cannot show the VLM above 4-bit or beyond a 16,384-token context: Stage II measures both.



Right clause cited, with 95% intervals. The lenient figure counts a clause of any of a board's defect types and is not a second headline.



Where a flagged board's time goes, at each stage's median.

Every figure: results/stage1/bench.json, written by `uv run spec-to-inspector bench` on 11 October 2026 under eval/PROTOCOL.md (written before the run; SHA-256 0109f982f5e9...); images a second: results/stage1/anomaly.json. Laptop: RTX 5070 Ti Laptop GPU, 12 GB; llama.cpp serving Qwen3VL-8B-Instruct-Q4_K_M.gguf with a 16,384-token context. Intervals are 95% Wilson intervals.

One codebase, two backends

CUDA on the laptop for Stage I; the Arc Pro B70 through Intel's own stack for Stage II. Which backend runs is a setting, not a code change.

	Stage I: the laptop MEASURED	Stage II: one Arc Pro B70 in the organiser's VM PLAN
CONSOLE	FastAPI and Jinja2, rendered on the server, nothing to compile · the corrections store is one SQLite file	
PIPELINE	parse → find → measure → name → decide → correct: one Python package, one command line, `spec-to-inspector`	
JUDGE CLIENT	one OpenAI-compatible client with a JSON-schema answer; the server's address and model are settings	
VLM SERVER	llama.cpp `llama-server` · Qwen3-VL-8B-Instruct at 4-bit (Qwen3VL-8B-Instruct-Q4_K_M.gguf) · 16,384-token context, one slot	Intel llm-scaler's vLLM image at a pinned tag, or upstream vLLM XPU · Qwen3-VL-8B-Instruct at FP16 or FP8 · 32,768-token context
ANOMALY MODEL	Anomalib 2.7.0 PatchCore on PyTorch 2.14.1+cu130, in FP32	the same Anomalib on PyTorch XPU in FP32, or exported to OpenVINO
PARSER	docling on the CPU	docling on the CPU · scanned pages through PaddleOCR-VL on the card
HARDWARE	NVIDIA GeForce RTX 5070 Ti Laptop GPU, 12 GB	ASUS UGen Intel Arc Pro B70, 32 GB

- **Why Qwen3-VL-8B-Instruct.** Intel's llm-scaler, "running on Intel Arc Pro B60 and B70 GPUs", lists `Qwen/Qwen3-VL-8B-Instruct` at FP16, FP8 and Int4; it is Apache-2.0; and llama.cpp serves the same model on the laptop, so both stages judge with one model.
- **Why upstream PyTorch XPU, OpenVINO and vLLM.** IPEX-LLM and Intel Extension for PyTorch are archived; their last commits are 28 January and 30 March 2026.
- **What is not yet shown.** We found no published figure for Anomalib or for any vision-language model on a B70. Stage II measures both; this deck quotes neither.

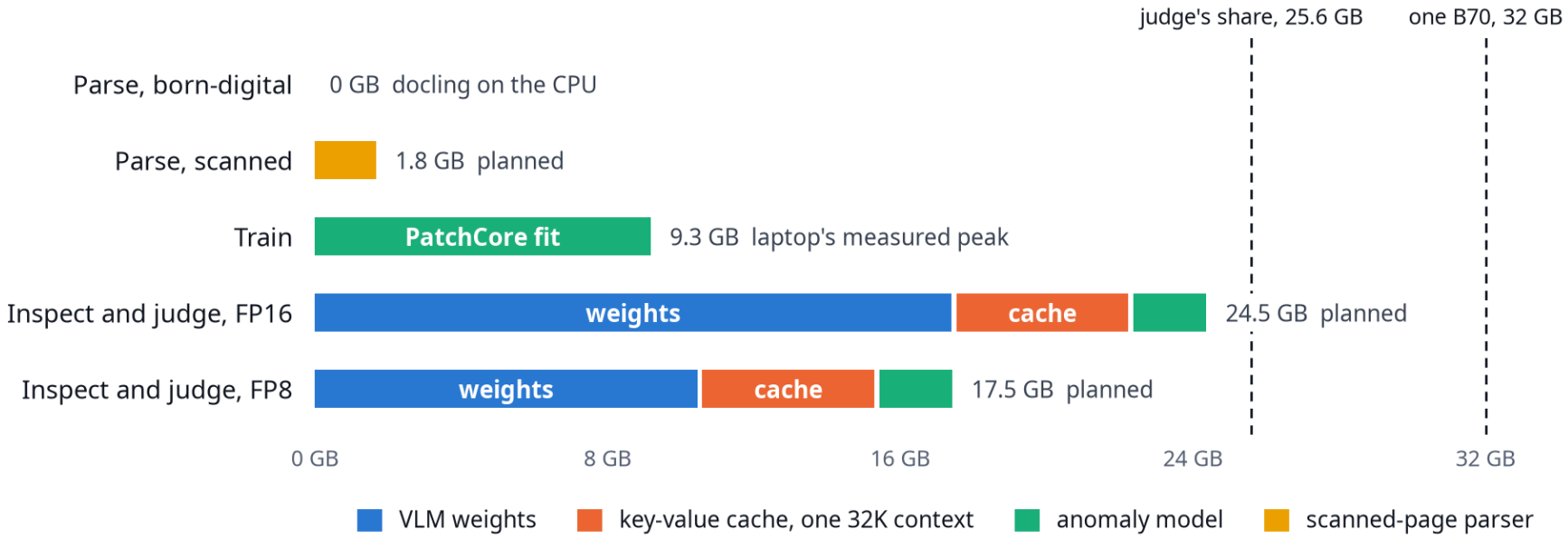
Stage I versions: results/stage1/anomaly.json and bench.json. llm-scaler's supported-model table and its 'running on Intel Arc Pro B60 and B70 GPUs': reference 11, read 11 October 2026. Archived repositories: reference 24, read 11 October 2026. The Stage II column is a plan; nothing in it has been run on a B70 by us.

One B70: the memory plan per stage

PLAN, TO BE MEASURED ON THE VM

The whole pipeline is planned to fit one Arc Pro B70's 32 GB, stage by stage, with the VLM at FP16 or FP8 rather than the laptop's 4-bit.

Our assumption about the Stage II environment. The organiser will provide a virtual machine on its SDV system from 26 October; its specification is not yet published. We plan for **one full Intel Arc Pro B70 (32 GB)** reached through a booked remote session, with no camera or USB device and limited outbound network. Every total on this page is planned from the vendors' file sizes and the model's configuration, and will be measured on the VM in week 1. If the VM exposes a share of a card rather than a whole one (the B70 supports up to four SR-IOV virtual functions), appendix C applies: the VLM at Int4, the stages strictly in turn, and the relevant clauses retrieved rather than the whole clause table held in context.



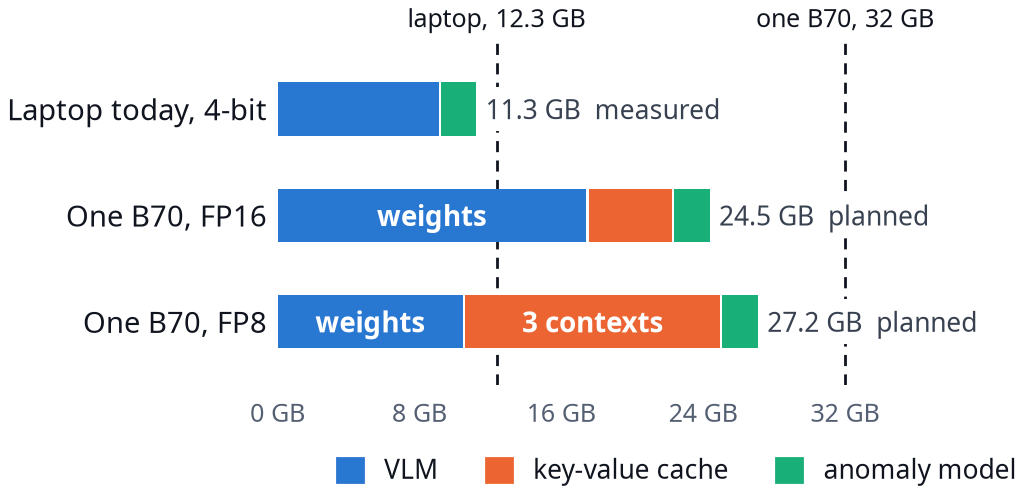
The weights are 17.53 GB at FP16 and 10.59 GB at FP8. The key-value cache is 147,456 bytes a token (36 layers, 8 key-value heads, head dimension 128, two bytes a value): 4.83 GB for one 32,768-token context. With vLLM's share of the card at 0.80 (25.6 GB), FP16 leaves 8.1 GB of cache, about 55,000 tokens, and FP8 15.0 GB, about 102,000. The other 6.4 GB is for the anomaly model, which peaked at 2.1 GB on the laptop while scoring. Training (9.3 GB reserved on the laptop) runs before the judge is loaded. Not counted: the vision encoder's working memory. GB is 10^9 bytes throughout.

Weight sizes and the cache's shape: Hugging Face file listings and config.json of Qwen/Qwen3-VL-8B-Instruct and its FP8 repository, read 11 October 2026 (reference 7), kept in deck/stage2-plan.json. Laptop peaks: results/stage1/anomaly.json and bench.json. Virtual functions: Intel, 'How Many SR-IOV Virtual Functions Are Supported on IBCs Intel Arc Pro B70 and B50 Graphics?', last reviewed 29 September 2026 (reference 23).

Why this needs the B70

One card's 32 GB holds an 8B vision-language model at FP16 or FP8, its context and the anomaly model together, on the supplier's premises, where a customer's confidential specification stays.

- **Precision.** The laptop's 12 GB holds the judge at 4-bit beside the anomaly model with little to spare (the card peaked at 11,395 MiB), and at 4-bit it cites the right clause on 38.5% of boards. Its weights are 17.53 GB at FP16, which no 12 GB card holds, and 10.59 GB at FP8, which leaves no room there for a context and the anomaly model. Whether precision closes the gap is the first thing Stage II measures.
- **Context, more than once.** One question is 4,683 tokens today, for a 6-page specification of 2,868 tokens; a 30-page one at that density is about 14,340. Every question in flight needs its own cache, 4.83 GB for a 32,768-token context. The laptop answers one at a time; the plan for the B70 at FP8 is 3 at once.
- **Everything resident.** The judge and the anomaly model stay loaded together through a batch, and training runs on the same card before them.
- **On premises.** A customer's specification is a confidential contract document. A card in the supplier's own workstation keeps it there; a cloud model does not.
- **The evidence that exists, and its limit.** Backend.AI measured 49.6 tokens a second on one B70 for "Qwen3 8B, 32K context, 4 concurrent": a text model, not a vision-language model. No vision-language figure on a B70 is published anywhere we found, so none is quoted here. An integrated GPU may hold an 8B model in shared memory; we assert no figure for it either. The B70's case is this workload at full precision and at line speed, and Stage II measures it.



Top bar: measured on the laptop (8,780 MiB for the judge, 2,028 MiB for the anomaly model). Lower bars: planned for one B70, to be measured on the VM.

Laptop figures: results/stage1/bench.json, spec.json and anomaly.json. B70 rows: planned from deck/stage2-plan.json (file sizes read 11 October 2026, reference 7); nothing here is measured on a B70. Backend.AI, 'Intel Arc meets Backend.AI', June 2026, table row 'Qwen3 8B, 32K context, 4 concurrent': reference 13, read 11 October 2026.

Stage II: the plan on the organiser's VM PLAN

Stage II replaces every laptop limit with a measured B70 figure, in four weeks of booked sessions, 6 November to 4 December.

WEEK 1 · 6 TO 12 NOVEMBER

Bring-up

Read the card the VM exposes (name, memory, whole card or virtual function) and record it. Install the pinned environment we carry in. Check PyTorch XPU and OpenVINO. Start the VLM server.

Output: the VM's measured shape, committed to the repository.

WEEK 2 · 13 TO 19 NOVEMBER

The anomaly model on the B70

PatchCore on PyTorch XPU in FP32, and through OpenVINO.

Output: training time, time per image and peak memory, against the laptop's 12.2 min, 86 ms and 8,003 MiB.

WEEK 3 · 20 TO 26 NOVEMBER

The judge at full precision

Qwen3-VL-8B at FP16 and at FP8 with the whole clause table in context.

Output: answer time, throughput, and clause citation on the same 200-defect set the laptop scored at 38.5%.

WEEK 4 · 27 NOVEMBER TO 4 DECEMBER

The full run

A 500-image batch end to end; memory per stage against page 14; the corrections loop across the batch.

Output: the 10-minute video recorded from the VM session; the repository with its diagram, README, model information and performance data.

It survives a booked session. Every step runs from files and resumes where it stopped: the pipeline needs no camera, no live feed and no network once installed, so a session is never lost to a reconnect.

What we need from the environment: one Arc Pro B70, a Linux or Windows 11 session, and a way to copy files in and out. We will fit this plan to the organiser's Computing Platform Guide when it is published.

THE LAPTOP'S LIMIT TODAY	MEASURED IN STAGE I	WHAT STAGE II PUTS IN ITS PLACE
The VLM's precision	4-bit, 38.5% right clause	FP16 and FP8, scored on the same boards (week 3)
Its context	16,384 tokens, one question at a time	32,768 tokens, and more than one question in flight (weeks 3 and 4)
The thresholds	chosen on the test split	chosen on a held-out split before the batch is scored (week 2)
The run	401 boards of two parts	a 500-image batch and a 200-defect clause set (week 4)

Dates: the contest's page, reference 1 (Stage II runs 6 November to 4 December 2026; booking of the organiser's environment opens 26 October), read 11 October 2026. Every output on this page is planned. The laptop figures they are set against: results/stage1/bench.json.

Expected outcomes: measurable targets, not promises

Stage II delivers a working inspector from a specification in minutes, a report that cites a clause on every reject, and an audit trail of overrides, each with a number beside it.

MEASURE	STAGE I, MEASURED ON THE LAPTOP	STAGE II ON THE B70 TARGET
Right clause cited, strict, the same 135 boards	**38.5%** (30.7% to 46.9%)	**70% or more.** What changes: FP16 and FP8 in place of 4-bit, and a good board's picture beside the crop for the judge to compare against
Defective boards stopped	188 of 200	no fewer, with the thresholds chosen on a held-out split
Good boards wrongly rejected	16 of 201, 8.0%	4% or less, on a held-out split
Boards referred to an engineer	14 of 200 defective, 9 of 201 good	reported as it falls: a referral is the honest outcome, not one to hide
Time for a flagged board	3.49 s median	measured at FP16 and FP8; no target, since no vision-language figure on a B70 is published to set one from
Specification to working inspector	12.3 min, one part	15 minutes or less, training on the B70
Peak GPU memory per stage	judge 8,780 MiB, anomaly model 2,028 MiB, training 8,003 MiB	within page 14's plan for 32 GB, measured
Size of the run	401 boards, two parts	a 500-image batch and a 200-defect clause set

Stage I column: results/stage1/bench.json (11 October 2026), laptop, VLM at 4-bit. The Stage II column holds targets and plans, not results: each is ours, set for this proposal, and none is a measured or published figure.

Who pays, and what it replaces

A supplier's quality department pays, per line or per specification onboarded, for a box on its own premises.

- **The buyer:** the quality department of a component or board supplier that inspects to its customers' specifications.
- **What it replaces:** weeks of rule-writing by hand or by a vision integrator for each new specification, and disputes argued from memory.
- **The shape:** an on-premises workstation with one ASUS UGen Intel Arc Pro B70, because specifications are confidential. Every licence is a hardware sale for ASUS and Intel.
- **The route:** the integrators who already sell inspection to these lines; the clause table is what they configure by hand today.
- **Why now:** a vision-language model that fits one workstation card is new, and Intel's own stack lists this one for it.
- **In the buyer's terms:** escape rate (defective boards that passed: 12 of 200 in Stage I), first-pass yield (good boards wrongly rejected: 16 of 201), and the engineer's hours returned to the line.

Before

Read the customer's specification | Write the inspection rules by hand | Tune them on the line

"It takes us weeks": the persona's words, not a measurement.

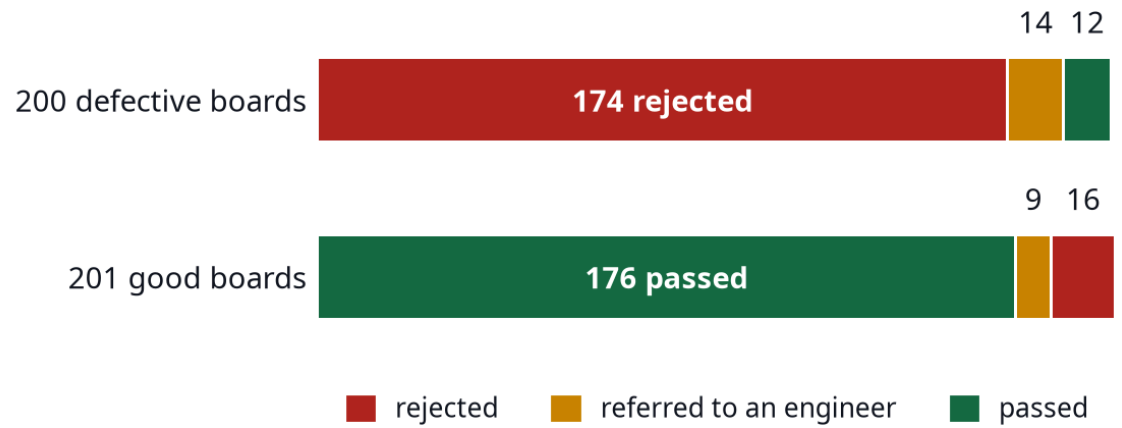
After

7.9 s parse the PDF into clauses

12.2 min train on good boards

The engineer reviews the clause table and the first batch: not timed

Measured on the laptop for one part of the demonstration specification.



What became of the Stage I test boards: the escapes and the false rejects a quality department counts.

The 'after' times: results/stage1/bench.json (11 October 2026), on the laptop. The 'before' is the persona's own words (page 3), not a figure. Escaped and wrongly rejected boards: the same file. No market-size or price figure is printed: none was found at a source.

Risks, and why the plan survives them

The risks are known, named and planned for; the measured ones are printed with their numbers.

RISK	WHAT THE PLAN DOES
The VM's shape is not published.	One whole B70 is assumed and said so (page 14); appendix C covers a share of a card and a Windows-only VM; the work is batches that replay from files.
Clause citation is weak at 4-bit: 38.5%.	Printed as measured. A wrong class or an unknown clause becomes "needs review", never a guess. Stage II measures FP16 and FP8 on the same boards, with a good board's picture beside the crop.
8.0% of good boards are rejected.	The judge clears few of the anomaly model's false alarms today, and we say so. Thresholds move to a held-out split; the engineer's "not a defect" is a correction like any other.
A correction acts as a rule about the class (14 of 19 control boards moved).	Corrections are scoped to the part, capped at three, and printed on every card they touched. Stage II measures narrower recall before any wider claim.
Measured extents are coarse.	The card prints what was measured and how; an area too small to outline is referred; Stage II raises the map's resolution.
The answer contract has a known defect: 3 of 200 defective boards referred as "malformed answer".	The cause is found: the model may write no digit, and stalls on a component's designator. The fix is the retry's wording; it moves at most those boards.
A specification written for public data looks staged.	It says so on its first page, in the console and here. The product is the clause schema and the citation, which a real specification fills the same way.
"How is this not Intel's defect explainer?"	Page 4: theirs explains a defect; ours binds each reject to the customer's own clause and limit, and keeps the engineer's override against it.
Booked hours on the VM run out.	Scripted runs, results saved per stage, nothing that needs a camera or a network once installed.

Measured figures: results/stage1/bench.json (11 October 2026) and corrections.json (11 October 2026). The assumption about the VM and its fallbacks: page 14 and appendix C.

References, and where to look

Every figure in this deck can be checked. Code and documentation: https://github.com/BrianIsaac/spec-to-inspector · Demo video: added before submission

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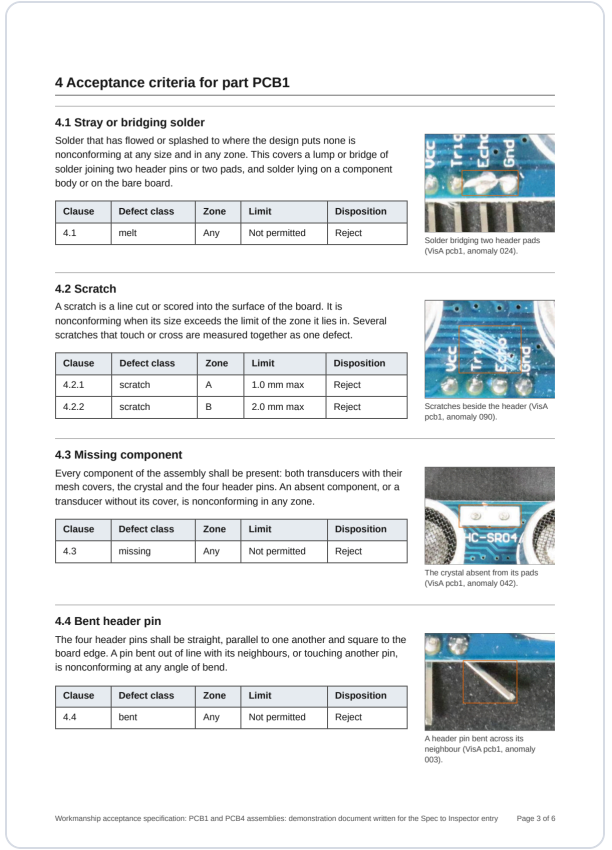
The customer's own specification, cited on every reject.

Each numbered address was fetched and answered on 11 October 2026 (`deck/build.sh --urls`). Measured figures are in results/stage1/ of the repository, each written by a command of the repository's own.

Appendix A: the demonstration specification

What the parser read: a 6-page workmanship specification for two boards, written for this entry in the style of a customer's document, and the 16 clauses it became.

CLAUSE	TITLE	PART	DEFECT CLASS	ZONE	LIMIT, AS PRINTED	PARSED	DISPOSITION	PAGE
4.1	Stray or bridging solder	PCB1	melt	any	Not permitted	any size	reject	3
4.2.1	Scratch	PCB1	scratch	A	1.0 mm max	> 1 mm	reject	3
4.2.2	Scratch	PCB1	scratch	B	2.0 mm max	> 2 mm	reject	3
4.3	Missing component	PCB1	missing	any	Not permitted	any size	reject	3
4.4	Bent header pin	PCB1	bent	any	Not permitted	any size	reject	3
5.1.1	Foreign material	PCB4	dirt	A	> 0.5 mm	> 0.5 mm	reject	4
5.1.2	Foreign material	PCB4	dirt	B	> 1.0 mm	> 1 mm	reject	4
5.1.3	Foreign material	PCB4	dirt	C	> 2.0 mm	> 2 mm	reject	4
5.2	Missing component	PCB4	missing	any	Not permitted	any size	reject	4
5.3	Extra component or object	PCB4	extra	any	Not permitted	any size	reject	4
5.4	Misplaced component	PCB4	wrong place	any	Not permitted	any size	reject	5
5.5	Damaged component or board	PCB4	damage	any	Not permitted	any size	reject	5
5.6.1	Scratch	PCB4	scratch	A	> 1.0 mm	> 1 mm	reject	5
5.6.2	Scratch	PCB4	scratch	B	> 2.0 mm	> 2 mm	reject	5
5.6.3	Scratch	PCB4	scratch	C	> 3.0 mm	> 3 mm	reject	5
5.7	Burn mark	PCB4	burnt	any	Not permitted	any size	reject	5



4 Acceptance criteria for part PCB1

4.1 Stray or bridging solder

Solder that has flowed or splashed to where the design puts none is nonconforming at any size and in any zone. This covers a lump or bridge of solder joining two header pins or two pads, and solder lying on a component body or on the bare board.

Clause	Defect class	Zone	Limit	Disposition
4.1	melt	Any	Not permitted	Reject

Solder bridging two header pads (VisA pcb1, anomaly 024).

4.2 Scratch

A scratch is a line cut or scored into the surface of the board. It is nonconforming when its size exceeds the limit of the zone it lies in. Several scratches that touch or cross are measured together as one defect.

Clause	Defect class	Zone	Limit	Disposition
4.2.1	scratch	A	1.0 mm max	Reject
4.2.2	scratch	B	2.0 mm max	Reject

Scratches beside the header (VisA pcb1, anomaly 090).

4.3 Missing component

Every component of the assembly shall be present: both transducers with their mesh covers, the crystal and the four header pins. An absent component, or a transducer without its cover, is nonconforming in any zone.

Clause	Defect class	Zone	Limit	Disposition
4.3	missing	Any	Not permitted	Reject

The crystal absent from its pads (VisA pcb1, anomaly 042).

4.4 Bent header pin

The four header pins shall be straight, parallel to one another and square to the board edge. A pin bent out of line with its neighbours, or touching another pin, is nonconforming at any angle of bend.

Clause	Defect class	Zone	Limit	Disposition
4.4	bent	Any	Not permitted	Reject

A header pin bent across its neighbour (VisA pcb1, anomaly 003).

Workmanship acceptance specification: PCB1 and PCB4 assemblies: demonstration document written for the Spec to Inspector entry Page 3 of 6

Page 3 of the document: three clauses of part PCB1, each with its table of limits and a photograph of the defect.

The clause table: specs/pcb-spec.clauses.json, as the parser wrote it from specs/pcb-spec.pdf (source: specs/pcb-spec.md). The page: page 3 of that PDF. Its photographs are from VisA (Amazon, CC BY 4.0, reference 6). The document is written for this entry: it is not a real customer's, its limits are illustrative, and it quotes no published standard.

Appendix B: how the figures were measured

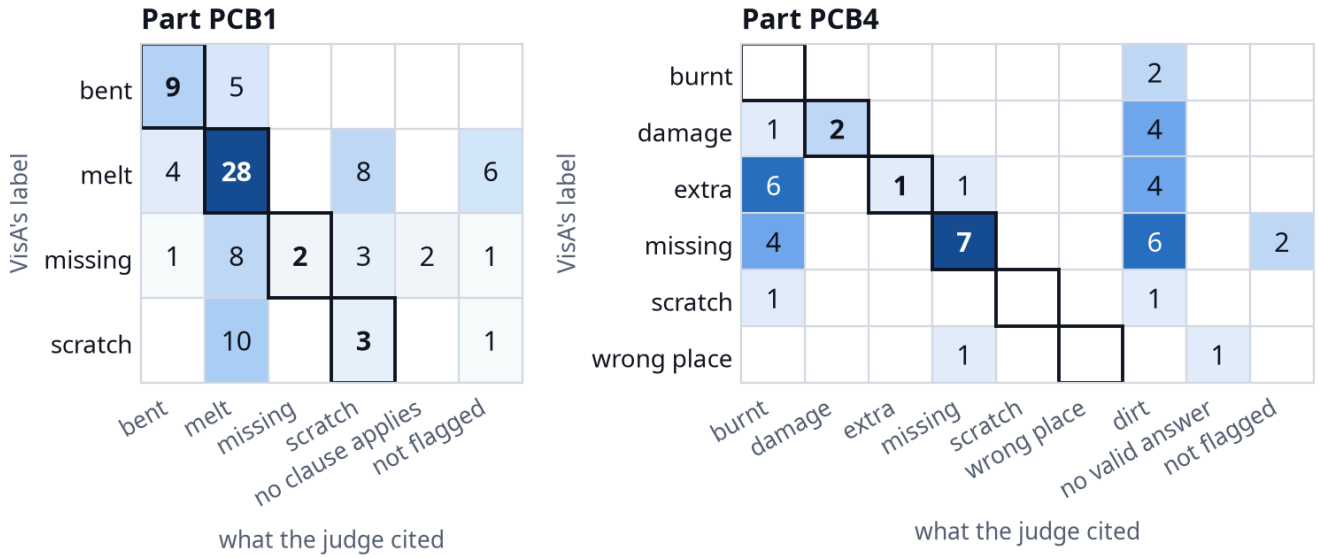
The commands, from a clean clone:

```
uv sync
uv run spec-to-inspector data fetch      # VisA's four PCB parts
uv run spec-to-inspector train --all     # one PatchCore model a part
scripts/serve_vlm.sh start               # llama.cpp, the 4-bit judge
uv run spec-to-inspector bench           # writes the bench record
scripts/serve_vlm.sh stop
```

The hardware: NVIDIA GeForce RTX 5070 Ti Laptop GPU, 12 GB; llama.cpp build b1-d812350 serving Qwen3VL-8B-Instruct-Q4_K_M.gguf with a 16,384-token context and one slot; PyTorch 2.14.1+cu130, Anomalib 2.7.0.

The labelled set: all 200 defective test boards of PCB1 and PCB4 in VisA's one-class split: 135 with one defect type, 65 with several. Nobody labelled a board for this entry. Each label is VisA's own annotation mapped to a clause by a table written before any model ran; the operator checked 20 rows chosen by a fixed rule and agreed with 20.

The protocol was written before the run. The headline is the clause cited on the most anomalous area against the expected clause, over the 135 single-defect boards; a board with no citation counts as wrong; multi-defect boards are scored apart; the corrections store is empty (0 of 401 decisions had a correction in the prompt); every question is asked cold. Run twice: 0 of 401 boards differed.



The headline's 135 boards: VisA's label down, what the judge cited across; the outlined diagonal is a right citation. The judge funnels PCB1 into "melt" and PCB4 into "dirt" and "burnt".

BESIDE THE HEADLINE	COUNT	SHARE (95% INTERVAL)
Lenient, multi-defect boards	38 of 65	58.5% (46.3% to 69.6%)
Per area, one defect type under it	60 of 148	40.5% (33.0% to 48.6%)
Single-defect boards flagged	125 of 135	92.6% (86.9% to 95.9%)
Defective boards referred	14 of 200	7.0%: not measured 8, malformed answer 3, outside every zone 2, no clause applies 1
Good boards flagged by the anomaly model	28 of 201	13.9% (9.8% to 19.4%)

Every figure: results/stage1/bench.json (11 October 2026); the protocol: eval/PROTOCOL.md (SHA-256 0109f982f5e9…), committed before the bench's code existed; labels: eval/clause_labels.csv and eval/clause_mapping.md; the spot-check: eval/spot-check-2026-10-11.md. Corrections: results/stage1/corrections.json. Anomaly model: results/stage1/anomaly.json.

Appendix C: if the VM is smaller than one B70 PLAN

The plan degrades in steps. It does not stop.

IF THE VM GIVES	WHAT CHANGES	WHAT IT COSTS
A share of a card: an SR-IOV virtual function. Intel states up to four on a B70; no size for one is published, and a quarter of 32 GB is the working guess	The VLM at **Int4**, which llm-scaler's table lists for Qwen3-VL-8B-Instruct, not FP8: FP8's 10.59 GB of weights does not fit a quarter of the card. The stages run strictly in turn. The context is shortened by retrieving the relevant clauses instead of holding the whole table	Accuracy: unknown until measured; the laptop's 4-bit figure, 38.5%, is the only evidence and is no promise for Int4. Time: the judge and the anomaly model are loaded and unloaded per batch. Retrieval adds a way to miss the right clause, which the bench would measure
A Windows-only VM	OpenVINO serves both models, the anomaly model and the VLM: llm-scaler is a Linux container. The pipeline, the console and the judge's client are unchanged	A second serving path to bring up in week 1; its speed and its precision options are measured there
No containers	The plain `uv` environment the laptop already uses; upstream vLLM XPU installed into it	Bring-up time in week 1
More than one card	The stages run side by side instead of taking turns	Nothing is lost

For scale, the laptop's own 4-bit files are 6.19 GB (weights and vision projector), and its judge peaked at 8,780 MiB with a 16,384-token context: a quarter of a B70 would be tighter than the laptop.

File sizes: Hugging Face listings read 11 October 2026 (reference 7), kept in deck/stage2-plan.json. Virtual functions: Intel's article, last reviewed 29 September 2026 (reference 23). llm-scaler's table: reference 11. The 4-bit clause figure: results/stage1/bench.json. Everything else on this page is a plan; no cost on it has been measured.

Appendix D: licences

Every part carries a licence that permits commercial use and redistribution, as the contest's consent form requires. Nothing non-commercial is in the build.

PART	WHAT IT IS HERE	LICENCE	WHERE IT RUNS
Spec to Inspector	this entry's own code	Apache-2.0	both stages
VisA, PCB1 to PCB4	the boards: training and test images	CC BY 4.0 (Amazon)	Stage I data
Anomalib 2.7.0	PatchCore, the anomaly model	Apache-2.0	both stages
`wide_resnet50_2` weights	PatchCore's backbone, from timm	Apache-2.0	both stages
PyTorch 2.14.1+cu130	the anomaly model's runtime	BSD-3-Clause	Stage I; PyTorch XPU in Stage II
docling 2.135.0	the specification parser	MIT	both stages
Qwen3-VL-8B-Instruct	the judge	Apache-2.0	4-bit in Stage I; FP16 or FP8 in Stage II
llama.cpp	the judge's server on the laptop	MIT	Stage I
Intel llm-scaler, vLLM	the judge's server on the B70	Apache-2.0	Stage II
PaddleOCR-VL-1.6	scanned specifications	Apache-2.0	Stage II
FastAPI, Jinja2, SQLite	the console and the corrections store	MIT, BSD-3-Clause, public domain	both stages

Licences as stated by each project's own page or model card (references 4 to 11); the PatchCore backbone's weights are those the training record names, results/stage1/anomaly.json. The repository's THIRD_PARTY.md is the full list, with each licence's URL.